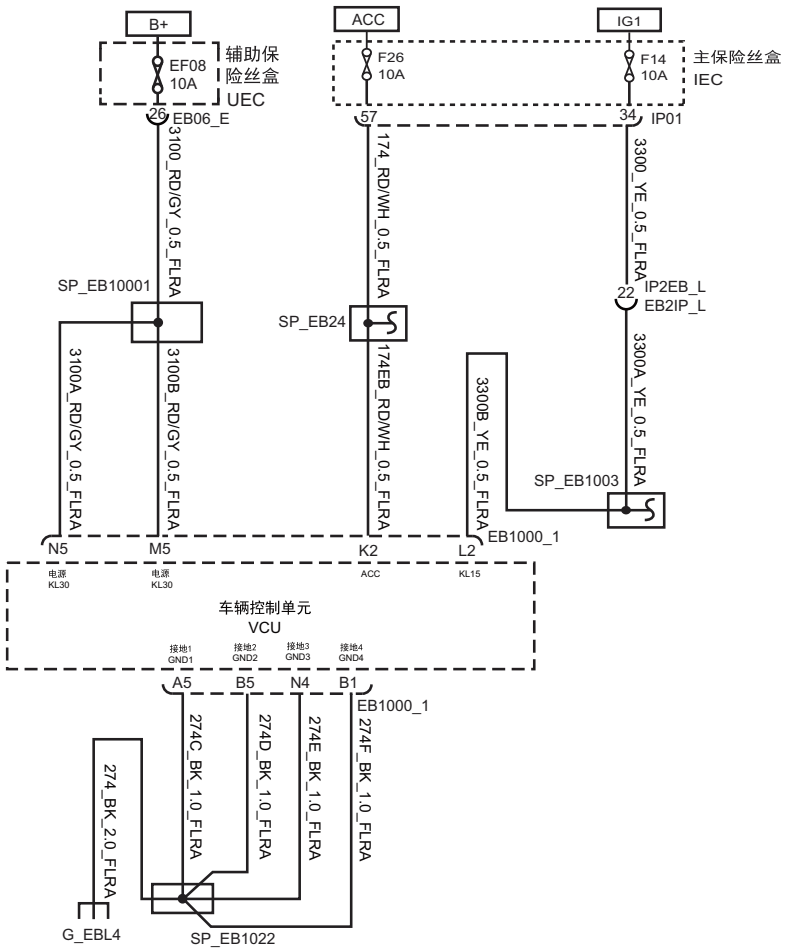
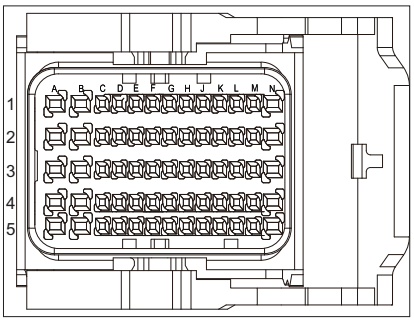
P0641、 P0651、 P0697

Fault diagnosis code
P0641：DEM_stU5V1SuplCfmd
P0651：DEM_stU5V2SuplCfmd
P0697：DEM_stU5V3SuplCfmd
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Circuit,terminal and adapting piece fault • Battery • VCU • Charging system • Insurance fuse
Conditions for fault code operation
点火开关处于 ON 位置。
Conditions for fault code setting
P0641
The sensor power supply 5V1 voltage is detected to be greater than 5.2V or less than 4.7V.
P0651
The sensor power supply 5V2 voltage is detected to be greater than 5.2V or less than 4.7V.
P0651
The sensor power supply 5V3 voltage is detected to be greater than 5.2V or less than 4.7V.
To set the effect of a fault code condition
Description of circuit diagram
The VCU will monitor whether all sensors and actuators are within normal range all the time. Meanwhile,it will check the connector for virtual connection, short circuit, open circuit and the connection rationality with other signals. When fault occurs, the VCU module will save the fault code corresponding to that fault and enable safety mode.

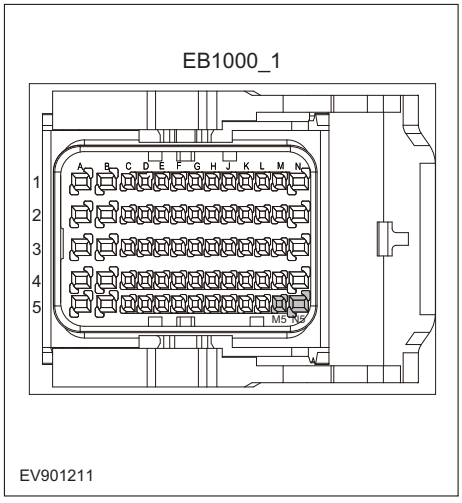
Circuit diagram

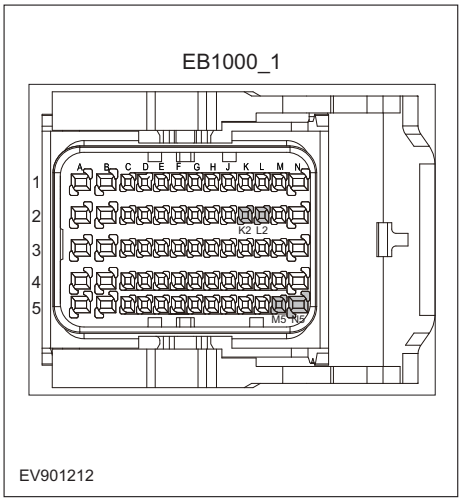
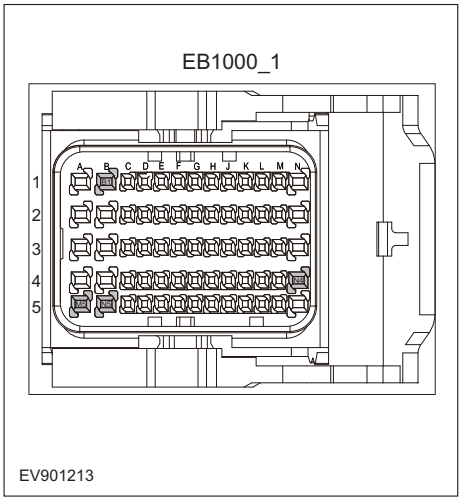


EB1000_1



EV901201

Troubleshooting steps		
Step 1	General inspection.	
		1 Check fuses EF08/F14/F26 for damage. 2 Check battery capacity. 3 Check the harness plug of air conditioning control module for damage, poor contact, aging and looseness. 4 Check whether the result is normal or not?
		NO ➞ Repair or replace the faulted parts.
YES ↓		
Step 2	Check fault codes.	
		1 Connect the fault diagnosis instrument to the diagnosis interface. 2 Turn the ignition switch to ON position. 3 Check the body control module with a diagnostic apparatus. 4 Is there any other fault code except for P0641/P0651/P0697?
		YES ➞ Refer to:DTC Summary list(VCU).
NO ↓		
Step 3	Check the power voltage of VCU battery.	
		1 Turn the ignition switch to OFF. 2 Disconnect the harness plug EB1000_1 of VCU. 3 Measure the voltage between the harness terminal of VCU and ground. Measuring circuit: voltage between terminal N5 on plug EB1000_1 and ground. Measuring circuit: voltage between terminal M5 on plug EB1000_1 and ground. **Standard value:10~14V** 4 Check whether the result is normal or not?
		NO ➞ Repair the circuit.
YES ↓		

Step 4	Check the power voltage when the VCU is started or operating
	1 Turn the ignition switch to ON. 2 Measure the voltage between the harness terminal of VCU and ground. Measuring circuit: voltage between terminal N5 on plug EB1000_1 and ground. Measuring circuit: voltage between terminal M5 on plug EB1000_1 and ground. Measuring circuit: voltage between terminal K2 on plug EB1000_1 and ground. Measuring circuit: voltage between terminal L2 on plug EB1000_1 and ground. **Standard value:10~14V** 3 Check whether the result is normal or not?
	NO → Repair the circuit.
YES ↓	
Step 5	Test whether the grounding circuit of VCU is open.
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug EB1000_1 of VCU. 3 Measure the resistance between the harness terminal of VCU and ground. Measuring circuit: resistance between terminal A5 on plug EB1000_1 and ground. Measuring circuit: resistance between terminal B5 on plug EB1000_1 and ground. Measuring circuit: resistance between terminal N4 on plug EB1000_1 and ground. Measuring circuit: resistance between terminal B1 on plug EB1000_1 and ground. **Standard value: < 1Ω** 4 Test whether the resistance is less than 1Ω?
	NO → Repair the circuit.
YES ↓	

Step 6	Test whether VCU is norm.
	1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug EB1000_1, EB1000_2 of VCU. 3 Check and repair. • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. • Fly out of pins-install new pins as needed. 4 Reconnect the VCU plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 5 Run the system and determine if the problem persists?
	YES → Replace the VCU.
NO ↓	
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems.	
Next Step ↓	
Diagnosis end.	